From Flat Fees to Verified Kilometres: An IoT-Based Revenue Redistribution Framework for Sustaining Off-Peak Private Bus Services in Sri Lanka

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***Abstract*—Public bus transit systems in developing regions face operational challenges, including poor service availability during low-demand periods, manual schedule management, untracked route deviations, unsafe driving behavior, and flat-rate permit fees that do not reflect actual usage. This paper presents a unified, IoT-enabled intelligent transportation framework integrating vehicle telemetry, map-matching verification, policy-based dynamic scheduling, distance-based revenue management, and real-time driver feedback. Onboard ESP32 microcontrollers with GPS modules and IMUs transmit location and motion data to a cloud platform via secure MQTT/HTTPS. The backend performs map-matching for route compliance, rule-based driver behavior classification, Peak/Normal/Night time-slot assignment, and automated distance-based permit fee calculation. Unlike conventional passenger-centric tracking systems, the framework introduces an economic governance mechanism that incentivizes operators to maintain off-peak services (9:00 PM–5:00 AM) through commissions linked to verified service delivery. Evaluation shows 96.2% accuracy in time-slot classification, 95.1% in policy enforcement, 91.6% - 93.6% detection accuracy for aggressive driving events, and reliable map-matched route tracking. Driver, owner, and regulatory interfaces give stakeholders operational visibility by connecting telemetry with scheduling policies, safety monitoring, and financial incentives – improving transit reliability, regulatory oversight, operator accountability, and public safety.**

***Keywords*—Intelligent Transportation Systems, Internet of Things, Driver Monitoring, Policy-Based Scheduling, Revenue Management, Map Matching, Smart Cities.**

I. Introduction

Public bus transit serves as the backbone of urban and suburban mobility in developing nations, offering affordable, accessible transport for millions of passengers. In Sri Lanka, private bus operators play a major role alongside state-operated services, particularly in densely populated regions such as the Western Province. Official statistics show a substantial private bus fleet operating across inter-provincial and provincial services [16], underlining the need for effective operational management. Despite this, the existing system continues to face challenges in service reliability, route compliance, passenger safety, and regulatory monitoring.

A major challenge arises during low-demand periods, particularly late at night and early morning, when passenger demand is often too low to give operators a strong incentive to maintain regular services on less profitable routes – leading to reduced frequency and inconsistent coverage. This is acknowledged by initiatives such as the Nisi Seriya night bus service, introduced to address the shortage of public transport during these hours. By the end of 2022, the National Transport Commission reported 216 Nisi Seriya services in operation [16], reflecting the continued importance of dedicated night transportation in Sri Lanka.

Maintaining such services, however, requires more than financial incentives or permits – authorities need reliable mechanisms to verify whether assigned services are actually carried out on their required routes and schedules. Conventional permit administration rarely provides continuous, real-time evidence of vehicle movement, leaving room for operators to cut services or deviate from assigned routes. The existing process – timetable determination, fee collection, permit issuance, and complaint monitoring – does not by itself give continuous, vehicle-level visibility.

A related limitation is the continued dependence on manual timetable and roadside monitoring, which is difficult to scale

and offers limited ability to respond quickly to breakdowns, delays, deviations, or service gaps. When data is collected manually or with delay, authorities may struggle to identify disruptions in time to arrange alternatives or adjust schedules. Real-time visibility of vehicle movement is therefore essential for accurate, responsive bus management.

Passenger safety is another major concern. Unsafe driving behavior – overspeeding, harsh braking, aggressive acceleration, sharp turning, and other abnormal patterns – increases risk to passengers and other road users while lowering service quality. Conventional supervision catches such behavior only through occasional inspections or post-incident investigations, whereas continuous onboard sensor monitoring can provide objective data to detect unsafe events and generate timely warnings.

Incentive-based initiatives also require reliable verification to ensure financial benefits are tied to actual service delivery. The Nisi Seriya program, for instance, provides concessions to encourage late-night and early-morning services, and official statistics confirm it remains active [16]. Without sufficient digital verification of vehicle movement and completed trips, it is difficult to establish a transparent link between performance and incentives – an opportunity to integrate real-time monitoring with automated financial and regulatory mechanisms.

To address these challenges, this paper proposes a unified IoT-enabled Smart Bus Route Monitoring and Revenue Management System. Rather than treating vehicle tracking, timetable management, driver monitoring, and financial administration separately, the framework brings them into a single closed-loop architecture, using real-time vehicle data as a common evidence layer for route verification, operational decisions, driver safety, and revenue calculation.

The proposed solution consists of four core components:

- IoT Data Acquisition and Map Matching: Continuously collects onboard telemetry – location and speed – and compares the trajectory against the authorized route to confirm compliance and service execution.
- Policy and Time-Slot Engine: Classifies operations into Peak, Normal, and Night periods, applies the corresponding policies, and supports dynamic responses to delays, breakdowns, and service gaps.
- Distance-Based Revenue Engine: Calculates permit charges from verified distance and applies incentives to operators maintaining required services during low-demand periods.
- Driver Monitoring and Mobile Interfaces: Uses real-time IMU data to detect unsafe driving and issue immediate warnings, while mobile interfaces deliver operational and safety information to operators and regulators.

Integrating these components lets the same verified data support multiple functions at once – for example, distance covered during a night service can verify route completion, confirm compliance, and calculate the incentive, while a route deviation or unsafe event can be logged for authority review.

This approach builds a continuous link between vehicle operations, regulatory policy, passenger safety, and financial management, reducing reliance on manual verification while maintaining a digital record of actual performance. Accumulated data can further help authorities identify recurring service gaps, inefficient routes, repeated safety violations, and areas needing improved coverage.

In summary, the proposed framework aims to shift bus management from a largely manual, reactive process toward a data-driven, continuously monitored, policy-aware system – improving service reliability, regulatory transparency, off-peak service delivery, and transportation safety in Sri Lanka.

II. RELATED WORK

Intelligent Transportation Systems (ITS) increasingly combine sensing, communication, positioning, and analytics to improve public transport monitoring. IoT lets vehicles continuously transmit operational data, while GPS-based AVL systems provide the foundation for real-time fleet monitoring [1]–[3]. Research has largely developed around vehicle tracking, scheduling, trajectory analysis, driver behavior monitoring, and performance-based management – mostly as separate domains.

A. IoT-Based Bus Monitoring and Real-Time Vehicle Tracking

GPS-based AVL is widely used for fleet management; early studies showed real-time location supports supervision, scheduling, and service improvement [2], [3]. Hounsell et al. showed AVL data supports passenger information and fleet operations while generating large volumes of historical data [3]. IoT architectures have since extended this by connecting sensors, networks, and cloud analytics, with surveys identifying transportation as a key domain for distributed data collection [1].

Most applications, however, only locate vehicles and estimate arrival time – location alone does not indicate whether a vehicle is following its authorized route, so additional spatial processing is needed for route-compliance information.

B. Map Matching and Route Compliance

Raw GPS trajectories cannot always be compared directly with road geometries, since positioning errors, signal loss, and dense urban environments introduce deviations. Map-matching addresses this by associating observations with the most likely road segments [4]. Quddus et al. reviewed geometric, topological, probabilistic, and filtering-based approaches, noting their limitations in complex environments [4].

Zhou et al. used historical bus locations and stop information to determine bus-line sequences through map matching [5], and trajectory-fitting approaches have similarly been proposed where existing maps fail to capture actual movement.

Map matching converts coordinates into route-level information but does not itself convert deviations into regulatory action. Here, it is treated as an evidence-generation mechanism for automated compliance verification.

C. Dynamic Scheduling and Real-Time Bus Operations

Traditional schedules are fixed in advance, yet operations are affected by congestion, demand, and delays. Research has explored real-time control – holding, rescheduling, stop-skipping, speed control – using location data to improve adherence. Fu and Yang demonstrated real-time bus-holding control, and later work examined prediction-based headway strategies [6], [7]. Berrebi et al. found real-time prediction improves headway stability, though effectiveness depends on prediction accuracy [7]. Gkiotsalitis identified holding, rescheduling, and stop-skipping as key approaches for disruptions [8].

These studies mostly target headway regularity. The proposed research extends this through a policy-based time-slot mechanism classifying operations into Peak, Normal, and Night periods, treating scheduling as input to policy enforcement rather than just a timetable parameter.

D. Sensor-Based Driver Behavior Monitoring

Driver behavior monitoring is central to ITS telematics. Accelerometers, gyroscopes, and GPS sensors capture motion linked to acceleration, braking, and turning. Ferreira et al. showed smartphone sensor data can classify driving events via machine learning [9], while Vlahogianni and Barmpounakis identified harsh longitudinal and lateral patterns using similar data [10]. Sarteshnizi and Khomeini showed accelerometer/gyroscope data can classify multiple maneuvers depending on configuration [11], and a systematic review confirms broad use of IMUs and GPS for detecting driving behavior [12].

However, these systems are typically evaluated as standalone safety applications – a detected unsafe event rarely influences route compliance or operator-level decisions. The proposed framework integrates driver events into the same infrastructure used for route and service monitoring.

E. Performance-Based Revenue and Incentive Mechanisms

Traditional payment and subsidy mechanisms struggle when profitability varies by route or period. Performance-based contract research proposes linking payments to measurable outcomes, aligning operator incentives with service objectives [13]. Ling et al. proposed a route-service-level bus-subsidy mechanism, showing that performance-linked incentives encourage operators to maintain service despite varying route economics [14]; later work continues exploring subsidy mechanisms tied to service-level indicators [15].

These studies support performance-based governance, but assessment typically relies on predefined indicators rather than continuous evidence. The proposed research instead introduces a distance-based revenue mechanism where verified movement directly feeds permit-fee calculation and incentive allocation.

F. Integrated Research Gap

ITS technologies are individually mature: GPS/AVL enables continuous visibility [2], [3]; map matching converts positioning into trajectories [4], [5]; real-time control supports schedule regularity [6]–[8]; sensor methods enable driver-behavior detection [9]–[12]; and subsidy research links financial mechanisms to outcomes [13]–[15]. Yet these remain standalone solutions – tracking focuses on location, control on regularity, monitoring on safety, incentives on performance – pointing to an opportunity for a unified architecture built on one stream of verified telemetry.

This research addresses that gap through an IoT-enabled smart bus monitoring and revenue management framework integrating four functions: (1) real-time telemetry and map-matched route verification, (2) policy-driven time-slot management, (3) distance-based permit-fee and incentive computation, and (4) real-time driver safety monitoring. Its contribution lies in linking physical vehicle evidence with policy and financial governance, tying performance directly to regulatory decisions – unlike systems offering only passenger information or isolated fleet monitoring, this framework closes the loop between vehicle operation, policy enforcement, safety monitoring, and economic incentives.

III. Methodology

The proposed solution follows a Design Science Research (DSR) approach [17] to design, implement, and evaluate an end-to-end IoT-enabled public bus monitoring and revenue management system. The methodology integrates real-time vehicle telemetry, route-compliance verification, policy-driven time-slot management, distance-based revenue calculation, and driver safety monitoring within a unified architecture, implemented through interconnected edge, cloud, backend, and client-side components that let bus-collected operational data support both real-time monitoring and regulatory decisions.

A. Overall System Architecture

The proposed system consists of four architectural layers: (1) IoT Field Layer, (2) Cloud Communication Layer, (3) Backend Processing Layer, and (4) Application Layer. Fig. 1 illustrates the overall architecture and data flow between bus-level sensing devices, cloud infrastructure, processing services, and stakeholder applications.

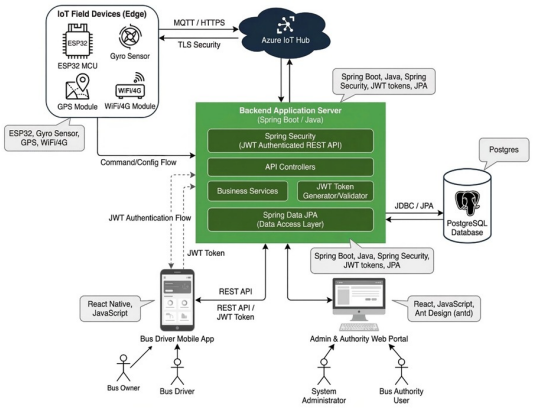


Fig. 1. Overall architecture of the proposed IoT-enabled smart bus monitoring and revenue management system.

At the IoT Field Layer, each bus carries an ESP32 microcontroller, a GPS module, and a 6-axis IMU. The GPS module provides position and speed data, while the IMU captures acceleration and angular-motion data for driver behavior analysis; the ESP32 handles sensor acquisition and packages readings into telemetry messages.

The Cloud Communication Layer secures connectivity between buses and the backend. Telemetry is transmitted via MQTT/HTTPS over TLS to Azure IoT Hub, which manages device communication and provides a scalable interface for telemetry and control data.

The Backend Processing Layer, built with Java and Spring Boot, hosts the core services – map-matching, policy and time-slot, revenue and incentive, and driver telematics engines. Spring Security and JWT handle authentication and authorization, while Spring Data JPA connects to PostgreSQL.

The Application Layer serves stakeholders: a React Native app lets drivers and owners authenticate, update trip status, report breakdowns, and receive safety alerts, while a React.js/Ant Design portal lets administrators monitor locations, inspect compliance, manage policies, and review safety and incentive data.

Data flow begins with sensor acquisition and continues through cloud transmission and backend processing, with the resulting information used for monitoring, policy evaluation, safety analysis, and financial calculation – linking physical vehicle activity to regulatory and economic decisions.

B. Real-Time Vehicle Tracking and Map-Matching Engine

The tracking module receives GPS telemetry from each vehicle through Azure IoT Hub. Each record contains the vehicle identifier, timestamp, geographic coordinates, and instantaneous speed. The reported trajectory is compared against the authorized route-a sequence of digitized road segments-using a simplified point-to-segment map-matching approach [4].

For a route segment between two points (x_1, y_1) and (x_2, y_2), the perpendicular distance from a GPS observation (x_t, y_t) to the line is calculated as:

$$d_{\perp} = \frac{|(y_2 - y_1)x_t - (x_2 - x_1)y_t + x_2 y_1 - y_2 x_1|}{\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}} \tag{1}$$

Coordinates are transformed into a suitable projected system before applying this calculation.

The result is compared against a predefined threshold θ_{route}. A deviation is flagged when the vehicle stays outside the route corridor for over 30 s. The event is stored in PostgreSQL and shown on the authority dashboard for monitoring and regulatory evaluation.

This mechanism converts raw GPS observations into route-level compliance information, providing an objective digital record of whether a vehicle follows its assigned route.

C. Policy and Dynamic Time-Slot Management Engine

The policy engine classifies each bus operation into one of three service periods: Peak, Normal, or Night. The classification is determined by the trip initiation time t_{start} and predefined time intervals configured by the transport authority:

$$\text{Service Slot} = \begin{cases} \text{Peak}, & t_{\text{start}} \in T_{\text{peak}} \\ \text{Normal}, & t_{\text{start}} \in T_{\text{normal}} \\ \text{Night}, & t_{\text{start}} \in T_{\text{night}} \end{cases} \tag{2}$$

where T_{peak}, T_{normal}, and T_{night} represent the corresponding configured operating intervals.

The assigned time slot determines the operational and financial policies applied to the trip. Night services between 9:00 PM and 5:00 AM are treated as off-peak and can qualify for incentives when the required route and service conditions are met.

The engine also processes operational disruptions reported through the mobile app or identified from telemetry. Events such as breakdowns and significant departure delays are evaluated against predefined policy constraints, and when a suitable standby vehicle is available, the system assigns it to maintain service continuity-responding to disruptions using current operational data rather than relying solely on manual intervention.

D. Distance-Based Revenue and Off-Peak Incentive Engine

The revenue engine links verified vehicle operation with permit-fee calculation. Rather than calculating permit liability solely from a fixed administrative amount, the mechanism uses the verified distance operated during each service period.

For a given billing period, the permit liability is calculated as:

$$R_{\text{permit}} = \sum_{k \in S} (D_k \times C_{\text{base}} \times \gamma_k) - I_{\text{night}} \tag{3}$$

where D_k is the verified distance for service period k, C_{base} is the base regulatory rate per kilometer, and γ_k is the policy multiplier for that period.

For qualifying night services, the incentive credit is calculated as:

$$I_{\text{night}} = D_{\text{night}} \times \mu_{\text{incentive}} \tag{4}$$

where D_{night} is the verified distance operated during the night period and $\mu_{\text{incentive}}$ is the incentive rate per kilometer.

The incentive is deducted from the operator's permit liability, linking financial benefits to verified delivery rather than operator declarations alone. The mechanism incentivizes essential services during low-demand periods while giving authorities a transparent basis for calculating charges and incentives.

E. Real-Time Driver Behavior Telematics

The driver telematics module uses onboard IMU data with GPS speed to identify predefined unsafe driving patterns, using tri-axial acceleration (a_x, a_y, a_z) and angular velocity $(\omega_x, \omega_y, \omega_z)$.

Four driving events are considered:

- **Overspeeding:** Speed v exceeds limit v_{limit} for at least 5 s.
- **Aggressive Braking:** Deceleration exceeds threshold θ_{brake}.
- **Aggressive Turning:** Yaw rate ω_z exceeds threshold θ_{turn} above a minimum operating speed.
- **Abnormal Slow Movement:** Speed stays below threshold $v_{\min}$ for at least 60 s with no identified congestion.

When detected, the system issues an immediate warning via the driver app and logs the event to the backend as part of the driver's safety history, later reviewable through the authority portal for performance monitoring and regulatory assessment.

F. Technology Implementation

The proposed framework combines embedded, cloud, backend, database, and client-side technologies. Each bus uses an ESP32 microcontroller with GPS and a 6-axis IMU for telemetry acquisition, communicating via MQTT/HTTPS over TLS to Azure IoT Hub. The backend, built with Java and Spring Boot, provides business logic and REST APIs secured through Spring Security/JWT, with PostgreSQL for data storage. Client-side, a React Native app serves drivers and owners, while a React.js/Ant Design dashboard serves authorities. This modular pipeline supports vehicle-level sensing through stakeholder-level monitoring, allowing components to be extended independently.

G. Experimental Evaluation

The prototype is evaluated using functional, accuracy, and operational measures across four areas: time-slot classification and policy enforcement, route-compliance detection, driver behavior detection, and telemetry processing.

Time-slot and policy performance are assessed by comparing assigned categories and decisions against ground-truth scenarios. Driver behavior detection uses controlled scenarios for overspeeding, aggressive braking/turning, and abnormal slow movement. Route-compliance performance uses authorized and deviated trajectories to test whether map matching correctly identifies violations.

Key metrics include classification accuracy, detection accuracy, false-positive rate, route-deviation detection, and processing latency, used to assess whether the framework provides reliable real-time information for monitoring, warnings, and regulatory decisions.

IV. RESULTS AND DISCUSSION

The prototype was evaluated through controlled field tests and simulated telemetry scenarios covering tracking, route compliance, time-slot management, revenue computation, and driver behavior monitoring, assessing the functional accuracy and real-time performance of all four core modules.

A. Real-Time Vehicle Tracking and Map-Matching

The tracking module was evaluated using 100 test observations for GPS positioning, speed monitoring, halt detection, and route-deviation detection. The results are summarized in Table I.

TABLE I
TRACKING AND MAP-MATCHING PERFORMANCE

Evaluation Parameter	Test Samples (n)	Achieved Result
GPS Positioning Error	100	4.2 m average
Speed Measurement Error	100	1.7 km/h average
Halt Detection	100	96.0% accuracy
Route Deviation Detection	100	94.95% accuracy
Average Telemetry Latency	—	1.4 s

The GPS subsystem achieved an average positioning error of 4.2 m and a speed measurement error of 1.7 km/h. Halt detection reached 96.0% accuracy, and route-deviation detection reached 94.95%. The average telemetry latency of 1.4 s indicates near-real-time monitoring capability.

B. Policy and Dynamic Time-Slot Management

The policy engine was evaluated using 500 trip records across the Peak, Normal, and Night periods, along with 30 test trips for delay estimation.

TABLE II
POLICY AND TIME-SLOT MANAGEMENT PERFORMANCE

Performance Parameter	Test Basis	Achieved Result
Time-Slot Classification	500 trip records	96.2% accuracy
Policy Rule Enforcement	Controlled scenarios	95.1% accuracy
Schedule Delay Estimation	30 test trips	0.5 min MAE
Standby Vehicle Assignment	Breakdown scenarios	< 3 s

The system achieved 96.2% time-slot classification accuracy and 95.1% policy enforcement accuracy. Schedule-delay estimation produced a mean absolute error (MAE) of 0.5 min. For simulated breakdown events, standby-vehicle assignment was completed in under 3 s, demonstrating a rapid automated operational response.

C. Distance-Based Revenue and Off-Peak Incentive

The revenue module was evaluated using verified distance records and predefined permit and incentive parameters.

TABLE III
REVENUE AND INCENTIVE COMPUTATION PERFORMANCE

Evaluation Parameter	Test Basis	Achieved Result
Distance Logging Verification	Reference distance records	99.1%
Permit Fee Computation	Validation cases	100%
Night Incentive Computation	Validation cases	100%
System Availability	72-h load test	99.4%

Distance logging achieved 99.1% agreement with reference records, while permit-fee and night-incentive calculations achieved 100% correctness across all validation cases. For qualifying services between 21:00 and 05:00, the system automatically calculated and applied the incentive credit to the operator's permit liability. These results validate the proposed distance-based revenue and incentive mechanism.

D. Driver Safety Telematics

The driver-monitoring module was evaluated using 30 controlled test instances for each of the four driving events defined in Section III-E.

TABLE IV
DRIVER BEHAVIOR DETECTION PERFORMANCE

Driving Event	Test Samples	Detection Accuracy
Overspeeding	30	93.3%
Aggressive Braking	30	91.6%
Aggressive Turning	30	91.8%
Abnormal Slow Movement	30	93.6%
Overall	**120**	**92.6%**

The module achieved detection accuracies ranging from 91.6% to 93.6%, with an overall accuracy of 92.6%. Abnormal slow movement recorded the highest accuracy, aggressive braking the lowest. Detected events were communicated immediately to the driver application and stored for authority-level safety monitoring.

E. Overall Discussion

The results demonstrate the feasibility of integrating vehicle telemetry, route compliance, policy management, revenue computation, and driver safety within a single IoT-enabled framework. The system achieved reliable tracking, automated service classification and disruption handling, verified distance-linked permit and incentive calculations, and consistent detection of predefined unsafe driving behaviors.

Overall, the framework establishes a closed-loop connection between vehicle operation, regulatory policy, driver safety, and financial incentives, providing a foundation for more accountable, responsive bus management.

V. CONCLUSION

This paper presented a low-cost, unified ITS framework connecting real-time vehicle telemetry with route compliance, policy enforcement, revenue management, and driver safety monitoring. The architecture integrates ESP32-based edge devices, Azure IoT Hub, a Spring Boot backend, and React-based mobile/web interfaces for centralized bus management.

Evaluation showed reliable performance across all functions. Tracking achieved 4.2 m average GPS error, 96.0% halt-detection accuracy, and 94.95% route-deviation accuracy. The policy engine achieved 96.2% time-slot classification accuracy and 0.5 min schedule-delay MAE, with standby-vehicle assignment under 3 s. The revenue module achieved 99.1% distance-logging verification and 100% correctness in permit and incentive calculations. Driver telematics achieved 92.6% overall detection accuracy.

The framework's key contribution is integrating vehicle monitoring with regulatory policy and financial incentives in a single closed-loop system, improving transit reliability, accountability, and public safety.

Future work will focus on edge-AI driver behavior classification, passenger-density estimation via additional sensing, and multi-route field evaluations assessing real-world performance, compliance, and long-term economic effectiveness.

REFERENCES

[1] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347–2376, 2015, doi: 10.1109/COMST.2015.2444095.

[2] A. X. Lobo, "A review of automatic vehicle location technology and its real-time applications," *Transport Reviews*, vol. 18, no. 2, pp. 165–191, 1998, doi: 10.1080/01441649808717009.

[3] N. B. Hounsell, B. P. Shrestha, and A. Wong, "Data management and applications in a world-leading bus fleet," *Transportation Research Part C: Emerging Technologies*, vol. 22, pp. 76–87, 2012, doi: 10.1016/j.trc.2011.12.005.

[4] M. A. Quddus, W. Y. Ochieng, and R. B. Noland, "Current map-matching algorithms for transport applications: State-of-the-art and future research directions," *Transportation Research Part C: Emerging Technologies*, vol. 15, no. 5, pp. 312–328, 2007, doi: 10.1016/j.trc.2007.05.002.

[5] Y. Zhou, G. Jiang, and G. Jiang, "Discover the road sequences of bus lines using bus stop information and historical bus locations," International Journal of Distributed Sensor Networks, vol. 15, no. 2, 2019, Art. no. 1550147719830552, doi: 10.1177/1550147719830552.

[6] L. Fu and X. Yang, "Design and implementation of bus-holding control strategies with real-time information," *Transportation Research Record*, vol. 1791, no. 1, pp. 6–12, 2002, doi: 10.3141/1791-02.

[7] S. J. Berrebi, E. Hans, N. Chiabaut, J. A. Laval, L. Leclercq, and K. E. Watkins, "Comparing bus holding methods with and without real-time predictions," *Transportation Research Part C: Emerging Technologies*, vol. 87, pp. 197–211, 2018, doi: 10.1016/j.trc.2017.07.012.

[8] K. Gkiotsalitis, "Bus operations optimization: A literature review on bus holding, rescheduling and stop-skipping," in *Proc. 99th Annual Meeting of the Transportation Research Board*, Washington, DC, USA, 2020.

[9] J. Ferreira Júnior, E. Carvalho, B. V. Ferreira, C. de Souza, Y. Suhara, A. Pentland, and G. Pessin, "Driver behavior profiling: An investigation with different smartphone sensors and machine learning," *PLOS ONE*, vol. 12, no. 4, p. e0174959, 2017, doi: 10.1371/journal.pone.0174959.

[10] E. I. Vlahogianni and E. N. Barmpounakis, "Driving analytics using smartphones: Algorithms, comparisons and challenges," *Transportation Research Part C: Emerging Technologies*, vol. 79, pp. 196–206, 2017, doi: 10.1016/j.trc.2017.03.014.

[11] I. T. Sarteshnizi and F. Tavakkoli Khomeini, "Sensitivity analysis of driving event classification using smartphone motion data: Case of classifier type, sensor bundling, and data acquisition rate," *Journal of Intelligent Transportation Systems*, vol. 28, no. 4, pp. 476–493, 2024, doi: 10.1080/15472450.2022.2140048.

[12] W. A. Al-Hussein, M. L. Kiah, and P. L. Yee, "A systematic review on sensor-based driver behaviour studies: Coherent taxonomy, motivations, challenges, recommendations, substantial analysis and future directions," *PeerJ Computer Science*, vol. 7, p. e632, 2021, doi: 10.7717/peerj-cs.632.

[13] D. Hensher and C. H. Mulley, "Performance-based quality contracts in bus service provision," *Transportation Research Part A: Policy and Practice*, vol. 37, no. 6, pp. 519–538, 2003.

[14] S. Ling, N. Jia, S. Ma, Y. Lan, and W. Hu, "An incentive mechanism design for bus subsidy based on the route service level," *Transportation Research Part A: Policy and Practice*, vol. 119, pp. 271–283, 2019, doi: 10.1016/j.tra.2018.11.006.

[15] Q. Luo, X. Bing, H. Jia, and J. Song, "An incentive subsidy mechanism for bus lines based on service level," Transport Policy, vol. 126, pp. 1–13, 2022, doi: 10.1016/j.tranpol.2022.07.006.

[16] National Transport Commission, *Transport Statistics / Private Bus Operational Data*, Ministry of Transport, Sri Lanka, 2022.

[17] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004, doi: 10.2307/25148625.